

Functional Annotation of *uvrD* (DNA helicase II, Q88C31 / PP_5352) in *Pseudomonas putida* KT2440

1. Summary / Answer to the Research Question

Q88C31 is UvrD (DNA helicase II), encoded by *uvrD*/PP_5352 in *Pseudomonas putida* KT2440. It is an ATP-dependent, superfamily-1A (SF1A) DNA helicase / single-stranded-DNA (ssDNA) translocase that moves along DNA with 3'→5' polarity and hydrolyzes ATP in a DNA-stimulated manner (EC 5.6.2.4). Its physiological substrate is **duplex chromosomal DNA bearing a 3' single-stranded loading region**, and it functions in the **cytoplasm at the nucleoid**. Mechanistically, UvrD unwinds/strips DNA at three converging genome-maintenance pathways: **(i) nucleotide excision repair (NER)** — displacing the lesion-containing strand after UvrABC incision; **(ii) methyl-directed mismatch repair (MMR)** — unwinding the error-containing strand after activation by MutL; and **(iii) recombination control** — acting as an **anti-recombinase** that removes RecA filaments from ssDNA and thereby tunes homologous recombination and the SOS response.

Identity is unambiguous, so function is confidently transferred from the extensively characterized *E. coli*/model-organism enzyme.

2. Gene / Protein Identity (Verification)

Property	Value
UniProt	Q88C31
Gene / locus	<i>uvrD</i> / PP_5352
Protein	DNA helicase II (DNA 3'–5' helicase II); EC 5.6.2.4
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125)
Length / mass	728 aa / 81.7 kDa
Domains	UvrD-like helicase ATP-binding (res 11–289) + UvrD-like C-terminal (res 290–567)
Motifs	Walker A / P-loop AGAGSGKT (res 32–39)
Family / orthology	Helicase SF1, UvrD subfamily; eggNOG COG0210 ; Pfam PF00580 + PF13361 ; InterPro IPR000212, IPR027417, IPR014016/17, IPR013986

Orthology check (this study): Domain architecture, gene symbol, and COG assignment match *E. coli* UvrD (P03018, 720 aa). A quick sequence comparison found **128 exact shared 5-mers (Jaccard $\approx$ 0.10)** between the two proteins — far above background for random proteins — and the canonical Walker-A P-loop (GSGKT) is present at the expected N-terminal position. The gene symbol "uvrD" is **not ambiguous** here: it corresponds to the classic, well-conserved bacterial DNA helicase II. Function can therefore be annotated with high confidence.

3. Primary Molecular Function — the Catalyzed Reaction and Substrate

UvrD is a **DNA-stimulated ATPase** that couples ATP hydrolysis to **directional 3'→5' translocation on ssDNA**, driving separation of duplex DNA (EC 5.6.2.4; GO:0043138 3'-5' DNA helicase, GO:0005524 ATP binding, GO:0016887 ATP hydrolysis).

- **Reaction (UniProt/RHEA):** $\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{phosphate} + \text{H}^+$ (RHEA:13065), and formally "couples ATP hydrolysis with the unwinding of duplex DNA by translocating in the 3'-5' direction." Under the 2021 EC revision, helicases were reclassified as **isomerases**/

translocases (EC 5.6.2.4): UvrD does not chemically modify DNA — it is a molecular motor that alters DNA conformation (duplex → two single strands).

- **Substrate specificity:** duplex DNA presenting a **3' single-stranded tail** for loading; the enzyme translocates on the tracking strand toward the duplex. Cofactor: **ATP + Mg²⁺**. Biochemically, the *Deinococcus radiodurans* ortholog is "a DNA-stimulated ATPase, can translocate along ssDNA in the 3'-5' direction and shows ATP-dependent 3'-5'... helicase activity"

(
P 24143224
).

- **Activation / regulation:** the **monomer is a processive 3'→5' translocase but a weak helicase**; efficient unwinding requires **activation by dimerization (via the 2B regulatory domain)** or by an **accessory partner protein**

(
P 42444602
);

P 38677494
).

This switch makes UvrD a *regulatable*, pathway-specific motor rather than a constitutively active unwindase.

3.1 Structural basis of the mechanism

UvrD-family helicases have a conserved **four-subdomain fold**: subdomains **1A and 2A** form the tandem RecA-like ATPase motor, while **1B and the mobile 2B domain** are accessory/regulatory (conserved in Q88C31: ATP-binding domain res 11–289 with Walker-A **AGAGSGKT**; UvrD_C domain res 290–567). Two RecA-like lobes hydrolyze ATP to ratchet along ssDNA; a **conserved β-hairpin "separation pin"** splits the duplex (its mutation abolishes helicase activity), and structures of the *D. radiodurans* ortholog "trap a DNA helicase undergoing a large-scale spiral movement around duplexed DNA"

(
P 24143224
).

The **2B domain is an allosteric switch**: it autoinhibits the monomer and is the general dimerization interface for the family, so activity is turned on by dimerization or by partner proteins

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P 42444602

P 29087333

This explains why UvrD is a *regulatable*, context-specific motor.

4. Biological Processes / Pathways

4.1 Nucleotide excision repair (NER)

After the UvrABC excinuclease dual-incises around a bulky/UV lesion, **UvrD unwinds the duplex 3'→5' to displace the lesion-containing oligonucleotide**, permitting resynthesis (Pol I) and ligation. "UvrD... plays an essential role in bacterial NER by unwinding the duplex DNA in the 3' to 5' direction to displace the lesion-containing strand"

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P 28120472

Consistent with this, *uvrD* loss confers **UV sensitivity**

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P 6355763

4.2 Methyl-directed mismatch repair (MMR)

UvrD is the **helicase that excises the error-containing strand** in MMR. **MutL activates UvrD** to "unwind DNA from the strand incision site up to and beyond the mismatch"

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P 38677494

Because the monomer is a poor helicase, MutL (or dimerization) is the required activator

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P 38677494

Loss of *uvrD* therefore produces a strong **mutator phenotype**, and *uvrD* is grouped genetically with *mutH/mutL/mutS* in the methyl-directed MMR pathway; the *uvrD* knockout is "mutator and UV-sensitive"

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P 6355763

This role directly maintains **replication fidelity and genome stability**.

4.3 Transcription-coupled repair (TCR) — proposed/debated

Beyond global-genome NER, an emerging model proposes that **UvrD backtracks RNA polymerase stalled at a lesion**, exposing the damage for NER, as an alternative to Mfd-mediated RNAP displacement: damage detection can occur "via transcription-coupled repair using UvrD-mediated backtracking or Mfd-controlled displacement of the RNA polymerase" (P 35235801).

In live *E. coli*, UvrD recruitment to DNA after UV damage is **strongly UvrA-dependent**, tying it to the NER machinery (P 38513450).

The relative contributions of the UvrD- vs Mfd-dependent routes remain under active debate.

4.4 Recombination control / SOS (anti-recombinase)

UvrD **removes RecA filaments from ssDNA** — "The UvrD helicase removes RecA filaments from RecA" (P 17364684).

— thereby limiting inappropriate homologous recombination and **modulating the SOS response** (e.g., *uvrD303* alleles decrease SOS expression; P 24084169).

uvrD is itself **DNA-damage-inducible under lexA-recA (SOS) control** (P 6355763).

integrating it into the DNA-damage response network. GO also annotates **recombinational repair** (GO:0000725).

5. Localization

UvrD is a **soluble cytoplasmic enzyme** acting on **chromosomal DNA at the nucleoid** (GO:0005829 cytosol; GO:0033202 DNA helicase complex; GO:0003677 DNA binding). The 728-aa sequence contains only the two intracellular helicase domains and a P-loop, with **no signal peptide or transmembrane segment** — it is neither secreted nor membrane-associated. It

engages DNA at replication forks and repair/recombination intermediates. (Localization is inferred from ortholog annotation and sequence features; no direct *P. putida* imaging is available.)

5.1 *P. putida*-specific functional network (STRING)

Independent of ortholog transfer, the STRING v12 interactome for PP_5352 places UvrD at a genome-maintenance hub, with top partners spanning every pathway above: **uvrB (0.934)**, **dnaB (0.906)**, **mutL (0.868)**, **uvrA (0.840)**, **uvrC (0.761)**, **dnaN β -clamp (0.733)**, **recA (0.717)**, **dinG (0.710)**, **mutY (0.668)**, **mfd (0.647)**, **recQ (0.641)**, **ruvA (0.621)** — i.e., NER excinuclease (UvrABC/Mfd), mismatch repair (MutL/MutY), the replisome (DnaB helicase, DnaN clamp), and recombination/fork processing (RecA, RuvA, RecQ, DinG). This corroborates the pathway assignments directly in *P. putida*.

6. Supported vs. Refuted Hypotheses

Supported: - H1 — Q88C31 is a *bona fide* UvrD ortholog (identity/domain/motif/COG evidence). ✓ - H2 — Primary function is ATP-dependent 3'→5' DNA helicase / ssDNA translocase. ✓ - H3 — Functions in NER (post-incision strand displacement). ✓ - H4 — Functions in MMR (MutL-activated; mutator on loss). ✓ - H5 — Acts as an anti-recombinase (RecA removal) within the SOS network. ✓ - H6 — Cytoplasmic, DNA-associated; not secreted/membrane-bound. ✓ - H7 — Activity is allosterically gated by the 2B domain (dimerization/partner activation); β -hairpin acts as the strand-separation pin. ✓ - H8 — In *P. putida* specifically, UvrD is networked with NER/MMR/replication/recombination machinery (STRING). ✓ - H9 — Possible role in transcription-coupled repair via RNAP backtracking (supported but debated). ●

Refuted / excluded: the gene symbol is **not** ambiguous or misassigned; no evidence supports an alternative (e.g., non-helicase) function.

7. Limitations and Future Directions

- Nearly all mechanistic and biochemical evidence derives from **orthologs** (*E. coli*, *D. radiodurans*, *Mtb* UvrD1, *P. falciparum*), not from *P. putida* PP_5352 directly; functional transfer relies on strong sequence/domain conservation.

- No *P. putida*-specific knockout, biochemistry, or localization study was located in this search; direct characterization (ATPase/helicase assays, $\Delta uvrD$ mutator/UV-sensitivity phenotyping, MutL/UvrABC interaction) would confirm the annotation.
- *P. putida* also encodes UvrD-family paralogs (e.g., Rep/PcrA-like helicases); partner specificity and any pathway division of labor in this organism remain to be established experimentally.

Evidence types used: protein-family/domain bioinformatics (UniProt, InterPro/Pfam, eggNOG-COG), sequence-motif analysis (Walker-A P-loop), *P. putida*-specific functional interactome (STRING v12), reaction/RHEA annotation (RHEA:13065, EC 5.6.2.4), structural/mechanistic studies (PMIDs 24143224, 42444602, 29087333), and primary/review literature on UvrD biochemistry and genetics (PMIDs 38677494, 28120472, 17364684, 24084169, 6355763, 35235801, 38513450, 34081984).

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